

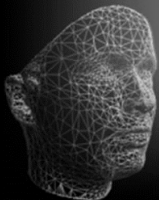
Advanced Topics in Computer Graphics II

Geometry Processing

Morphable Models



January 28, 2025





Problem: Given a large set of similar shape specimens. How to efficiently describe the shape variation for editing and fitting?

How to define a simple mathematical model of body shape? It should

- ▶ look like real people
- ▶ move like real people
- ▶ be low-D, differentiable, have joints, be easy to animate and to fit to data
- ▶ be compatible with standard graphics tools.



Human Body Shapes

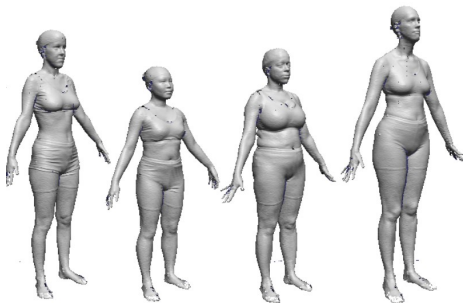


In this lecture: Human Body Shapes.



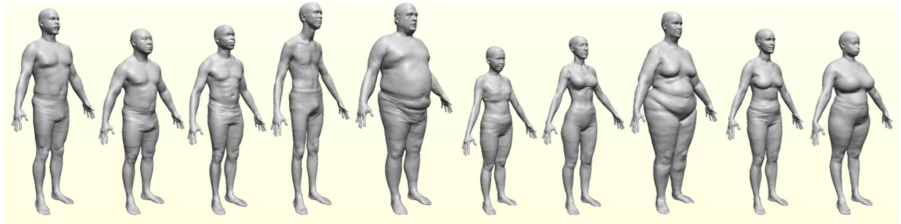
Cyberware 3D whole body scanner

1998-2001: Acquisition of **CAESAR** dataset



Generate 3-D surface data to revise current anthropometric databases of civilian males & females ages 18-65 of various weights in the U.S. and Europe.

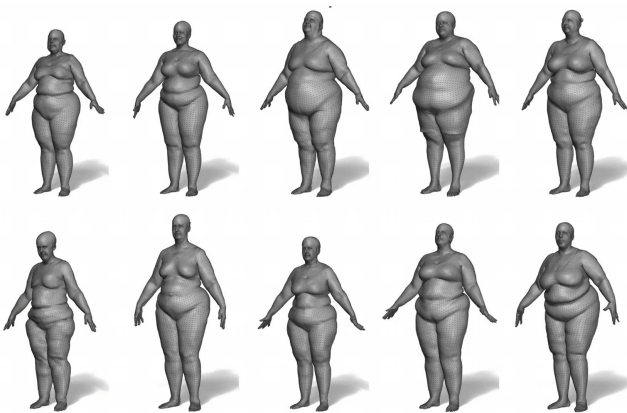
B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594



Examples from the CAESAR dataset (Civilian American and European Surface Anthropometry Resource Project)



CAESAR dataset (Civilian American and European Surface Anthropometry Resource Project)



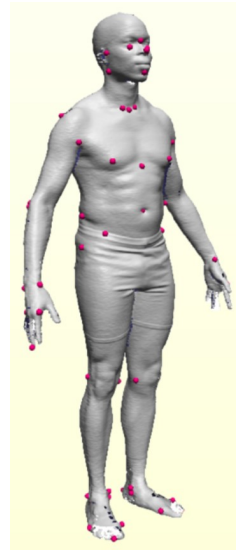
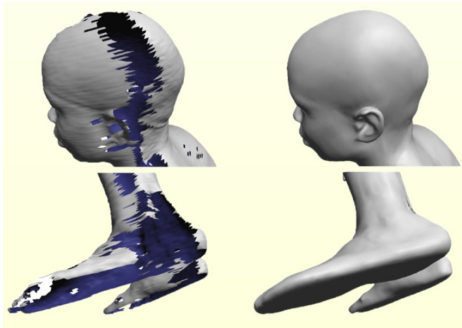
(taken from N. Mahmood et al. *SMPL made Simple*. Tutorial at CVPR 2021. June 2021. URL: <https://smpl-made-simple.is.tue.mpg.de/>)



Modeling the space of human bodies: data set



- ▶ 125 male and 125 female scans
- ▶ 74 white markers were placed on the subject at anthropometric landmarks
- ▶ demographic data such as age, weight, and ethnic group were recorded





- ▶ Fit the scanned surface to a template surface T
- ▶ Affine transformation T_i per vertex v_i
- ▶ Objective function

$$E = \alpha E_d + \beta E_S + \gamma E_m$$



$$E_d = \sum_{i=1}^n w_i d(T_i v_i, D)^2$$

Data error: template and target surface as close as possible

1

$$E_S = \sum_{(i,j) \in E} \|T_i - T_j\|_F^2,$$

where $E = \{(v_i v_j) \mid \in \text{edges}(T)\}$

Transformation smoothness

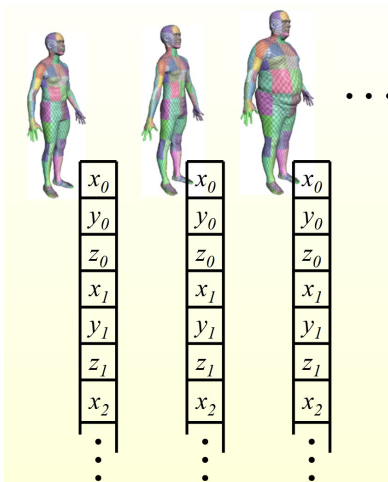
$$E_m = \sum_{i=1}^m \| (T_{\kappa_i} V_{\kappa_i} - m_i) \|^2$$

Marker error

¹B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594.



Modeling the space of human bodies: Data alignment



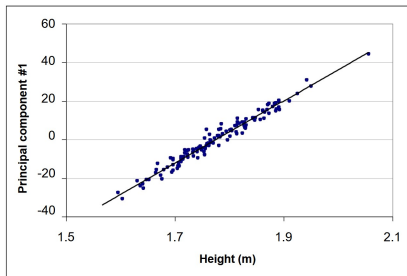
**Statistical analysis
(PCA) on vertex positions**



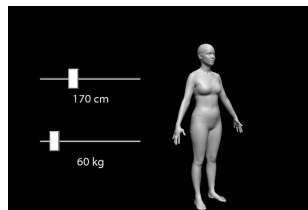
Average male



Statistical analysis (PCA) on vertex positions



from³



³B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594.

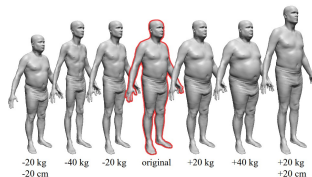
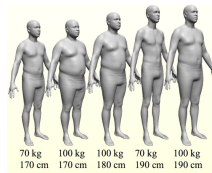


Statistical analysis (PCA) on vertex positions (from⁴)

- Feature based synthesis
- Learned mapping between controls and the PCA coefficients

$$M[f_1, \dots, f_l, 1] = p,$$

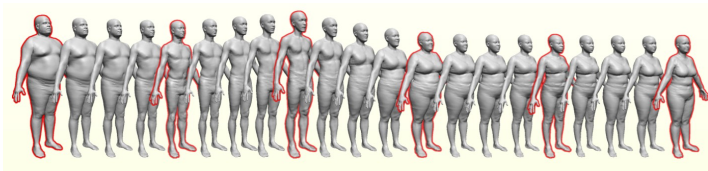
where f_i are the feature values of an individual, and p are the corresponding PCA weights.



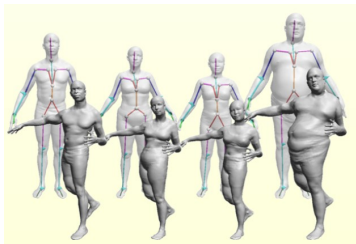
⁴B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594.



Morphing



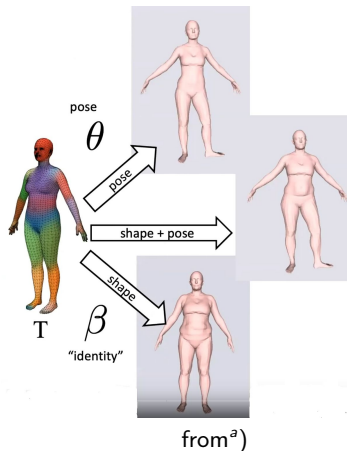
Skeleton transfer



from B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594



- ▶ Previous method: all humans in neutral poses
- ▶ Scape decouples deformation into rigid and non-rigid components
 - ▶ Body shape variation model and pose deformation model
- ▶ Dataset
 - ▶ 70 poses of a particular person in wide variety of poses
 - ▶ 37 scans of different people in neutral poses + CAESAR
- ▶ Reconstruct skeleton (16 parts) for each pose
- ▶ Compute Delta-‘PCA’-like blendshapes over triangles wrt. skel.



(taken

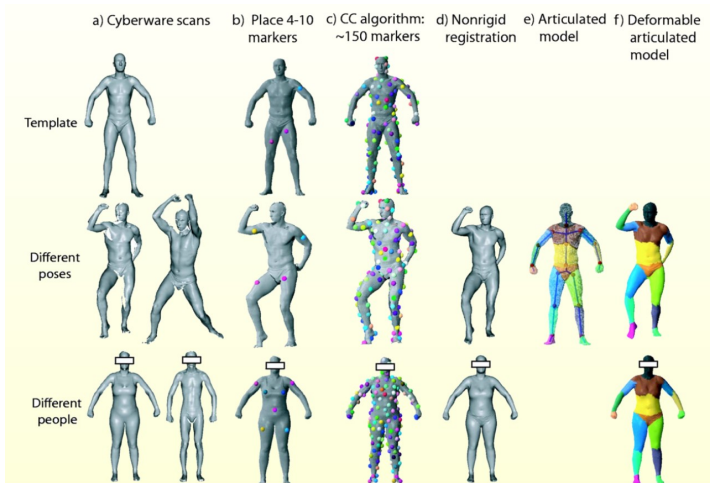
from^{a)}

^aN. Mahmood et al. *SMPL made Simple*. Tutorial at CVPR 2021. June 2021. URL: <https://smpl-made-simple.is.tue.mpg.de/>.

⁵D. Anguelov et al. "Scape: shape completion and animation of people". In: *ACM SIGGRAPH 2005 Papers*. 2005, pp. 408–416.

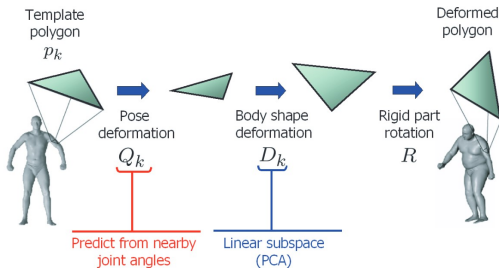


Scape processing pipeline



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⁶Image credit: D. Anguelov



- ▶ Transformations are represented as 3×3 matrices
- ▶ Order of application preserves proper deformation scaling.
- ▶ The model correlates the values of matrices Q_k to the nearby joint angles, and learns a low-dimensional linear manifold for the values of matrices D_k .

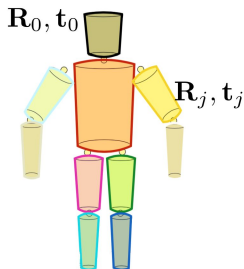
(image taken from⁷)

- ▶ Template triangles p_k get transformed to generate a new human shape instance.
- ▶ Accounts for several sources of triangle deformation:
 - ▶ non-rigid muscle deformations Q_k due to changes in pose,
 - ▶ non-rigid body shape deformations D_k accounting for changes between different humans
 - ▶ and articulated part rotations R .

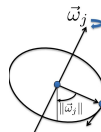
⁷D. Anguelov. "Learning models of shape from 3D range data". PhD thesis. Citeseer, 2006.



Parameterize every body part separately ?



$$X_{pose} = \{R_0, t_0, \dots, R_N, t_N\}$$

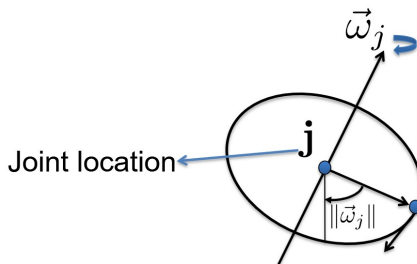


- ▶ Rotations are composed of 9 numbers
- ▶ 6 additional constraints to ensure that the matrix is orthonormal
- ▶ suboptimal for optimization
- ▶ Describe rotations with **Exponential Maps** Let $\hat{\omega}_j$ be the crossproduct matrix of the normalized ω_j then

$$R = \exp(\|\omega_j\| \hat{\omega}_j) = I + \sin(\|\omega_j\|) \hat{\omega}_j + (1 - \cos(\|\omega_j\|)) \hat{\omega}_j^2$$



The transformation associated with a rotational joint is



9

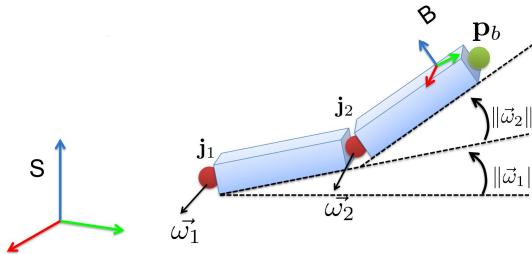
Rigid Motion at joint j

$$G(\omega, j) = \begin{pmatrix} [\exp(\|\omega\|\hat{\omega})]_{3 \times 3} & j_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{pmatrix}$$

⁹Image credit: G.Pons-Moll



Example of Kinematic Chain

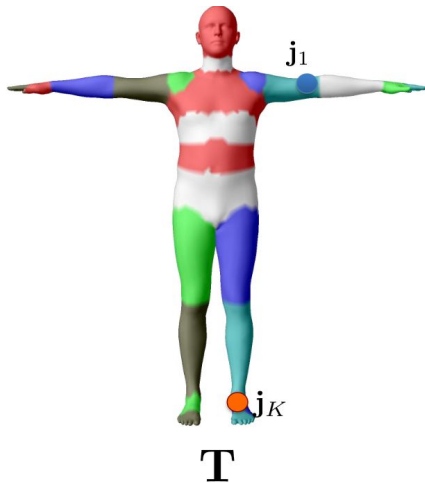


10

The coordinates of the point in the spatial frame are:

$$p_s = G(\omega_1, \omega_2, j_1, j_2)p_b = G(\omega_1, j_1)G(\omega_2, j_2)p_b$$

¹⁰Image credit: G.Pons-Moll



Given a set of joint locations

$$\mathbf{J} = (j_1, \dots, j_K)^t$$

The pose is defined as the vector of concatenated part axis-angles

$$\boldsymbol{\theta} = (\omega_1, \dots, \omega_K)^t$$



Factorize into joint rotation R_k^i and triangle transformation Q_k^i

$$\arg \min_{Q_1^i, \dots, Q_P^i} \sum_k \sum_{j=2,3} \|R_k^i Q_k^i \hat{v}_{k,j} - v_{k,j}^i\|^2 + w_S \sum_{k_1, k_2 \text{ adj}} I(l_{k_1} = l_{k_2}) \cdot \|Q_{k_1}^i - Q_{k_2}^i\|^2$$

Joint rotations R_k^i

Learn how to **predict** Q_k^i from the twists of its adjacent joints



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¹¹Image credit: D. Anguelov

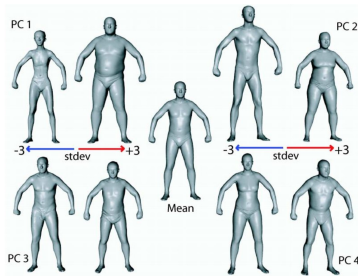
Complete deformation model is given by

$$v_{k,j}^i = R_{l[k]} D_k^i Q_k^i \hat{v}_{k,j},$$

where D_k^i describe the **Body Shape deformations**

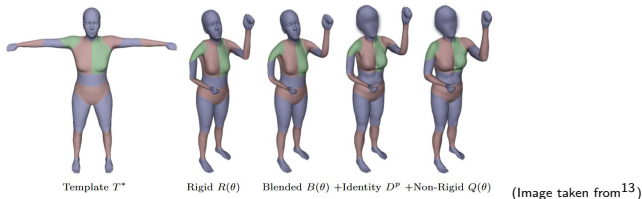
Body shape deformations are given by PCA: **Covariance matrix EV** and **mean**

$$D^i = U\beta^i + \mu$$





The Blend Scape Model

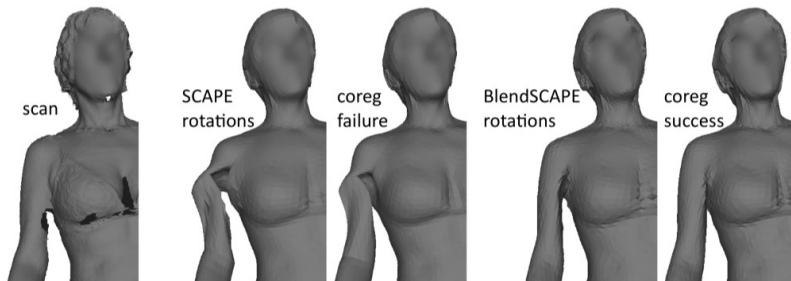


- ▶ Since each static part of the template is independently rotated, the final stitched body surface can collapse, crease or fold near joints
- ▶ Q can be trained to correct these artifacts but if these artifacts are sufficiently severe they can cause convergence problems during coregistration.
- ▶ Use **linear blend of rotations**

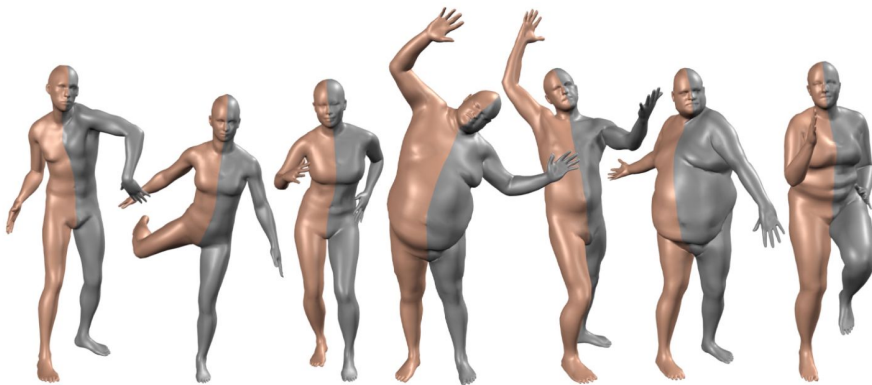
$$B_k(\theta) = \sum_i w_{k_i} R^i(\theta),$$

where θ be a vector collecting all the relative rotations and $R(\theta)$ represent the absolute rotations of the triangles in a part.

¹³D. A. Hirshberg et al. "Coregistration: Simultaneous alignment and modeling of articulated 3D shape". In: *Computer Vision—ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part VI* 12. Springer. 2012, pp. 242–255.



- ▶ SCAPE vs BlendSCAPE: Rigid part rotations, R , can introduce significant artifacts early in the fitting process, particularly when scan noise (e.g. holes) coincides with part boundaries of the model. Coregistration has difficulty eliminating these artifacts.
- ▶ In contrast, B from the BlendSCAPE model introduces smoother, less significant artifacts, which coregistration rapidly corrects.



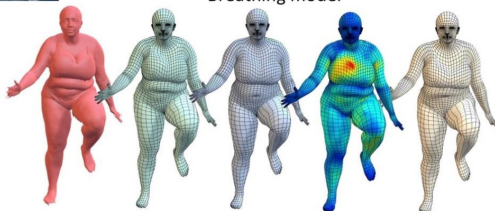
Paper: M. Loper et al. “SMPL: A skinned multi-person linear model”. In: *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*. 2023, pp. 851–866



Breathing model



4D scanner (60 fps)



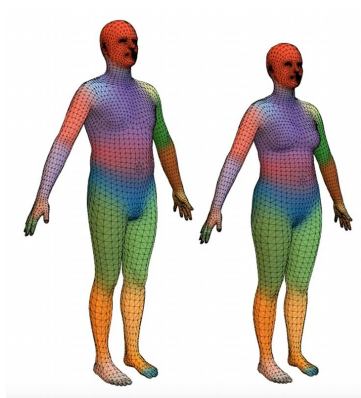
4D scan

4Cap

BlendSCAPE

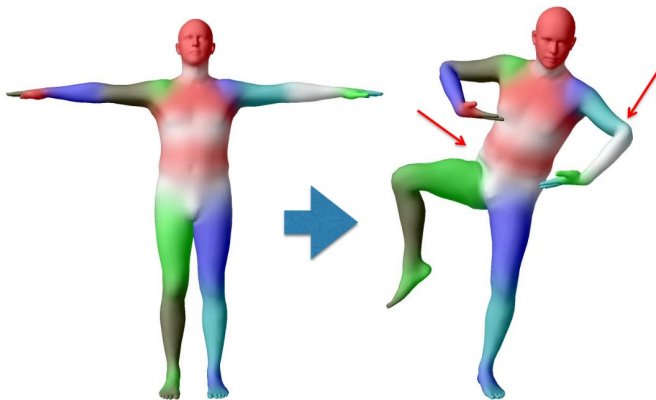
Soft tissue

Dyna



- Represent the whole body as a neutral gender 3D mesh.
- SMPL uses about $N = 7000$ vertices in 3D, $\mathbf{T} \in \mathbb{R}^{3N}$ (Template mesh)
- Mesh is segmented into parts that can rotate in a kinematic tree with blend weights $\mathcal{W} \in \mathbb{R}^{N \times K}$ with K joint positions $\mathbf{J} \in \mathbb{R}^{3K}$.
- Skinning function:

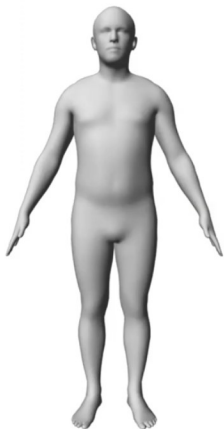
$$W(\mathbf{T}, \mathbf{J}, \mathcal{W}, \theta) \mapsto \text{vertices}$$



$$\bar{t}'_i = \sum_{k=1}^K \omega_{k,i} G'_k(\boldsymbol{\theta}, \mathbf{J}) \bar{t}_i$$

Vertices are a linear combination of transformed template vertices. ¹⁵

¹⁵Image credits: G. Pons-Moll





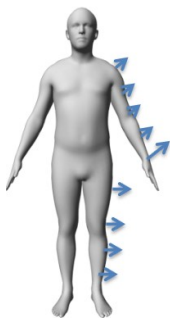
A **blend shape** is a vector of **vertex displacements** in a rest pose





A **blend shape** is a set of vertex displacements in a rest pose

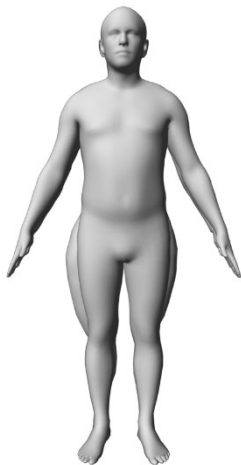
Pose blend shapes: correct for LBS problems



$$\mathbf{P} = \text{vec}\left(\begin{bmatrix} \Delta x_1 & \Delta y_1 & \Delta z_1 \\ & \vdots & \\ & \vdots & \\ \Delta x_N & \Delta y_N & \Delta z_N \end{bmatrix}\right) \xrightarrow{\text{Offset 1}} \in \mathbb{R}^{3N}$$



With blend shape corrections



Pose blend shapes: correct for LBS problems ¹⁹



Standard skinning $W(\mathbf{T}, \mathbf{J}, \mathcal{W}, \vec{\theta}) \mapsto \text{vertices}$

²⁰Image credits: Michael Black



Standard skinning $W(\mathbf{T}, \mathbf{J}, \mathcal{W}, \vec{\theta}) \mapsto \text{vertices}$

SMPL model

$M(\vec{\theta}, \vec{\beta}) = W(\mathbf{T}_F(\vec{\beta}, \vec{\theta}), \mathbf{J}(\vec{\beta}), \mathcal{W}, \vec{\theta}) \mapsto \text{vertices}$

²⁰Image credits: Michael Black



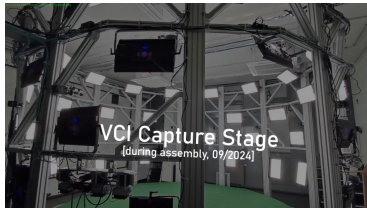
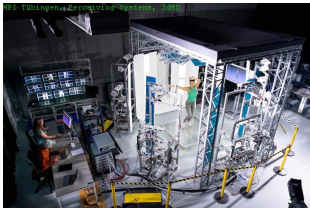
Standard skinning $W(\mathbf{T}, \mathbf{J}, \mathcal{W}, \vec{\theta}) \mapsto \text{vertices}$

SMPL model

$M(\vec{\theta}, \vec{\beta}) = W(\mathbf{T}_F(\vec{\beta}, \vec{\theta}), \mathbf{J}(\vec{\beta}), \mathcal{W}, \vec{\theta}) \mapsto \text{vertices}$

SMPL is skinning parameterized by pose $\vec{\theta}$
and shape $\vec{\beta}$

Capturing the body surface in 4D

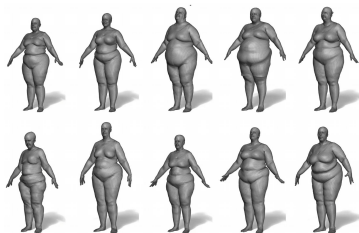


Databases: Collect 3D scans from thousands of people

- ▶ Caesar: Generate 3-D surface data to revise current anthropometric databases of civilian males & females ages 18-65 of various weights in the U.S. and Europe.
 - ▶ Collecting new body measurements
 - ▶ 2,500 U.S. & 2,500 European civilians from 1998-2001
 - ▶ Collect, analyze, and report new database in a format useful to scientists & engineers around the world
- ▶ SMPL, A Skinned Multi-Person Linear Model
 - ▶ **Paper:** M. Loper et al. "SMPL: A skinned multi-person linear model". In: *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*. 2023, pp. 851–866
 - ▶ Statistical human shape spaces learned from the meshes fitted to the pre-processed CAESAR body scans
 - ▶ Comes with example FBX files with animated SMPL models, and code for using SMPL in Python, Maya and Unity.

Additional Databases

- ▶ MPII Human Shape, Version 1.0,
 - ▶ Statistical human shape spaces learned from the meshes fitted to the pre-processed CAESAR body scans
 - ▶ **Paper:** L. Pishchulin et al. "Building Statistical Shape Spaces for 3D Human Modeling". In: *Pattern Recognition* (2017)
- ▶ FAUST Dataset (MPI Tübingen)
 - ▶ FAUST contains 300 real, high-resolution human scans of 10 different subjects in 30 different poses, with automatically computed ground-truth correspondences.
 - ▶ **Paper:** F. Bogo, J. Romero, M. Loper, et al. "FAUST: Dataset and evaluation for 3D mesh registration". In: *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*. Piscataway, NJ, USA: IEEE, June 2014
- ▶ DFAUST Dataset (MPI Tübingen)
 - ▶ Dataset of high-resolution 4D scans of human subjects in motion, captured at 60 fps.
 - ▶ **Paper:** F. Bogo, J. Romero, G. Pons-Moll, et al. "Dynamic FAUST: Registering Human Bodies in Motion". In: *IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*. July 2017



(taken from N. Mahmood et al.
SMPL made Simple. Tutorial at
CVPR 2021. June 2021. URL:
<https://smpl-made-simple.is.tue.mpg.de/>)

- **Paper:** D. Anguelov et al. "Scape: shape completion and animation of people". In: *ACM SIGGRAPH 2005 Papers*. 2005, pp. 408–416
- A model parametrizes "deviations" from a template mesh.
- Simplifies learning and inference.



- [1] B. Allen, B. Curless, and Z. Popović. “The space of human body shapes: reconstruction and parameterization from range scans”. In: *ACM transactions on graphics (TOG)* 22.3 (2003), pp. 587–594.
- [2] D. Anguelov. “Learning models of shape from 3D range data”. PhD thesis. Citeseer, 2006.
- [3] D. Anguelov, P. Srinivasan, D. Koller, S. Thrun, J. Rodgers, and J. Davis. “Scape: shape completion and animation of people”. In: *ACM SIGGRAPH 2005 Papers*. 2005, pp. 408–416.
- [4] F. Bogo, J. Romero, M. Loper, and M. J. Black. “FAUST: Dataset and evaluation for 3D mesh registration”. In: *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*. Piscataway, NJ, USA: IEEE, June 2014.
- [5] F. Bogo, J. Romero, G. Pons-Moll, and M. J. Black. “Dynamic FAUST: Registering Human Bodies in Motion”. In: *IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*. July 2017.



- [6] D. A. Hirshberg, M. Loper, E. Rachlin, and M. J. Black. “Coregistration: Simultaneous alignment and modeling of articulated 3D shape”. In: *Computer Vision—ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part VI 12*. Springer. 2012, pp. 242–255.
- [7] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. “SMPL: A skinned multi-person linear model”. In: *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*. 2023, pp. 851–866.
- [8] N. Mahmood, T. Bolkart, A. A. A. Osman, J. Tesch, D. Tzionas, and M. J. Black. *SMPL made Simple*. Tutorial at CVPR 2021. June 2021. URL: <https://smpl-made-simple.is.tue.mpg.de/>.
- [9] L. Pishchulin, S. Wuhrer, T. Helten, C. Theobalt, and B. Schiele. “Building Statistical Shape Spaces for 3D Human Modeling”. In: *Pattern Recognition* (2017).